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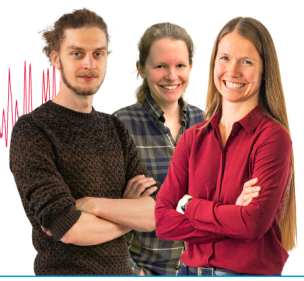
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STATIS Methodology: An Overview

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Abstract. The STATIS (Structuration des Tableaux a Trois Indices de la Statistique) methodology is a data analysis technique which computes Euclidean distances between studies of the same observations obtained in k different circumstances. The Escoufier operators are used to obtain a geometrical representation for the studies in a series. In this study we present developments in the STATIS methodology, namely how to make inferences when for each treatment of a model we have a series of studies.

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Introduction

The Analyse Conjointe de Tableaux (STATIS method Structuration des Tableaux a Trois Indices de la Statistique) is an exploratory technique of multivariate data analysis based on linear algebra and especially on Euclidean vector spaces. This method was introduced by [3], and developed by several authors, in particular by [4] and [5]. STATIS can be used to analyze data in a variety of fields, including agriculture, environmental science, sociology, and psychology. It is a powerful tool for exploring complex data structures and identifying relationships between multiple variables, but it is important to be cautious about the results and to interpret them in the context of the data and the assumptions of the methodology. The central idea of this methodology is to compare configurations of the same individuals or variables in different circumstances or moments in time. Therefore, this method allows us not only to identify the moments in time in which the most significant changes occurred, but also to rank variables and individuals according to their contributions to the total changes. We are also able to draw the trajectory of each individual or variable around its compromise (average) position. In the STATIS methodology we study series of studies. The studies are matrix triplets
$(\mathbf{X}_i, \mathbf{D}_{p_i}, \mathbf{D}_n)$, where the first matrix is a data matrix where columns contains the values of the considered variables in the considered objects. They are two types of series:

1. The objects are the same in all studies. We then have $n \times p_i$ data matrices $\mathbf{X}_i$ with n the number of objects and p_i the number of variables in the r-th study, $i = 1, \ldots, k$. In each study we have also matrices, in general diagonal, $\mathbf{D}_{p_i}$ and $\mathbf{D}_n$ with weights of the variables in the r-th study, $i = 1, \ldots, k$, and of the objects in the series. In this case we have the triplet $(\mathbf{X}_i, \mathbf{D}_{p_i}, \mathbf{D}_n)$, $i = 1, \ldots, k$.
2. The variables are the same in all studies. The data matrix $\mathbf{X}_i$ being of type $n_i \times p$ with n_i the number of objects in the r-th study and p the number of variables in the series. We have weights $\mathbf{D}_p$ and $\mathbf{D}_{p_i}$ for variables and objects. In this case we have the triplet, $(\mathbf{X}_i, \mathbf{D}_{p_i}, \mathbf{D}_{n_i}), i = 1, \ldots, k$.

[2], defined the matrices

$$\mathbf{A}_i = \mathbf{X}_i \mathbf{D}_{p_i} \mathbf{X}_i^t \mathbf{D}_n, \ i = 1, \ldots, k$$

and the matrices

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$$\mathbf{B}_i = \mathbf{X}_i^t \mathbf{D}_{n_i} \mathbf{X}_i \mathbf{D}_p,\ i = 1, \ldots, k$$

to represent series of studies of the first and series of studies of the second type, respectively. The STATIS methodology defines

$$S_{ij} = tr(\mathbf{A}_i \mathbf{A}_j^t),\ i = 1, \ldots, k,\ j = 1, \ldots, k$$

as the interstructure matrix with elements being the Hilbert–Schmidt scalar products $(A_i|A_j) = tr(\mathbf{A}_i \mathbf{A}_j^t),\ i = 1, \ldots, k$, for series of first type and $(B_i|B_j) = tr(\mathbf{B}_i \mathbf{B}_j^t),\ i = 1, \ldots, k$, for series of second type.

The symmetric matrix shus the sprecteal decomposition

$$\mathbf{S} = \sum_{j=1}^{k} \lambda_j \boldsymbol{\alpha}_j \boldsymbol{\alpha}_j^t$$

with $(\lambda_j, \boldsymbol{\alpha}_j),\ j = 1, \ldots,\ j = 1, \ldots, k$ been the pairs of it's eigenvalues and eigenvectors. The h-th components of vectors $\lambda_j \boldsymbol{\alpha}_j, h = 1, \ldots, k$ are the coordinates of the point representing the h-th study. There is a common structure when these points lie along the first axis which happens when the first eigenvalue is dominant (e.g. [4]).

Models and inference

If we have a common structure, $\lambda_1, \ldots, \lambda_j,\ j = 1, \ldots, k$ will be small, $\lambda_l << \lambda_j,\ l = 2, \ldots, k$, and the points representing the studies fall along the first axis, this led [7] to assume, the model

$$\mathbf{S} = \lambda \boldsymbol{\alpha} \boldsymbol{\alpha}^t + \bar{\mathbf{E}}$$

with

$$\bar{\mathbf{E}} = \frac{1}{2}(\mathbf{E} + \mathbf{E}^t),$$

when $\bar{\mathbf{E}}$ is such that

$$\mathbf{Vec}(\mathbf{E}) \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_{k_2}).$$

We can now use λ_1 and $\boldsymbol{\alpha}_1$ the first eigenvalue and eigenvector of matrix $\mathbf{S}$ as estimators of λ and $\boldsymbol{\alpha}$. Actually through simulations the good behaviors of those estimators had already been showed (e.g. [8]). This model enables the concentration of the information, concentrated in matrix $\mathbf{S}$, into the estimated structure vector

$$\tilde{\boldsymbol{\beta}} = \lambda_1 \boldsymbol{\alpha}_1$$

and the sum of squares of residuals,

$$V = \|S\|^2 - \|\boldsymbol{\beta}\|^2.$$

Moreover, $\tilde{\boldsymbol{\beta}}$ is the estimator of the structure vector

$$\tilde{\boldsymbol{\beta}} = \lambda \boldsymbol{\alpha}.$$

Validating the model is centered on establishing the predominance of the first eigenvalue of $\mathbf{S}$. Actually, confidence intervals for

$$\tau = \frac{\lambda^2}{\sigma^2}$$

(see [8]). In [1], the models formulation was refined and the common structure of degree 1, associated to a single dominant eigenvalue for matrix $\mathbf{S}$, has been extended for h degree common structure

$$\mathbf{S} = \frac{h}{2} \lambda_i \boldsymbol{\alpha}_i \boldsymbol{\alpha}_i^t + \mathbf{E} = \sum_{i=1}^{h} \boldsymbol{\beta}_i \boldsymbol{\alpha}_i^t + \mathbf{E}$$

with $\mathbf{E}$ an covariance matrix and

$$\boldsymbol{\beta}_i = \lambda_i \boldsymbol{\alpha}_i,\ i = 1, \ldots, h.$$

The structure vectors $\boldsymbol{\beta}_i,\ i = 1, \ldots, h$, were estimated and the sum of squares of residues was obtained.

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Structure families of series of studies

Up to now we have considered a single series of studies. Now we assume having for each treatment of a model such a series. Our goal will be to study the action of the factors in that model on the structure vectors. Two types of analysis are considered:

- Transversal Analysis on the homologous components of the structure vectors. Usually only the first such vectors are considered;
- Longitudinal Analysis on linear applications on the estimated structure vectors. This type is now adequate when these is a progressive in the studies of the series. For instance if they were carried out in successive times. As stated above these analysis are usually carried out for the first structure vectors since, many times, they are the sole ones for which we have estimators with sufficient precisions.

Several applications are presented has led to other useful approaches to data analysis ([1]; [7]). [1] consider the validation of these model studied of local Portuguese elections of years (1985, 1989, 1993, 1997 and 2001). In this study, four districts stretching from North to South along the coast and four others along the border with Spain were selected. For each of these districts, two townships were selected. One of them contained the district capitalcity. Thus we had good coverage of the territory, being able to apply the proposed model to the 16 series of studies. That model was based on a new dimension for modelling the inter-structure of a series of studies in the ACT method.

Final remarks

STATIS methodology is used to identify patterns or structures in the data and to explore the relationships between multiple variables. Although to have a solid algebraic formulation, the STATIS methodology only gives empirical conclusions. The development of models has allowed for the possibility of making inference about the relationships between variables and to test hypotheses about the underlying structures in the data. This provides a deeper understanding of the data and allows for the development of predictions and insights about the population and can help researchers to make more informed decisions and to further advance their understanding of the relationships between the variables.

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